

27 B

Using Einstein's Photoelectric equation, $\frac{hc}{\lambda} = \phi + eV_s$

$$\begin{aligned}\phi &= \frac{hc}{\lambda} - eV_s \\ &= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{150 \times 10^{-9}} - (1.60 \times 10^{-19})(2.1) \\ &= 9.9 \times 10^{-19} \text{ J} \\ &= 6.2 \text{ eV}\end{aligned}$$

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